

PHASE 1:

AI COUNSELLING APP FOR HIGH-STRESSED STUDENTS

A Jungian Cognitive Approach to Emotional Support

University: IU International University

Course: DLBCSPJWD01

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BACKGROUND

Modern technology is evolving at an exponential pace, and with it, so is the pressure we place on ourselves. We now live in a world more fast-paced than any generation before us. Students, particularly those in Asia, carry an enormous emotional burden when preparing for exams, a kind of accumulated emotional debt that rarely gets acknowledged, let alone addressed.

I am not an exception to this. Drawing from my own experience and years of expertise building full-stack EdTech applications, I want to create something that genuinely helps. This app is designed to stand on its own or work alongside any existing study or wellness application. It acts as a quiet companion that shows up when students need to be heard the most.

TARGET AUDIENCE

Students that are stressed during the exam preparation phase.

BENEFITS

At the heart of this project is a simple belief: a calm mind is the foundation of everything, in exams and in life.

After using this app, students will walk away feeling heard in a way they may never have experienced before. This will not happen through generic advice or automated responses, but through something that feels almost magical in its attentiveness. That sense of being truly understood is what clears the mental fog and gives students the emotional readiness to face whatever journey lies ahead.

WHAT MAKES MY APP DIFFERENT

Most people think of technology as a cold, number-crunching tool, and technically, they are not wrong. But as a developer, my job goes beyond making something that works. My job is to make something that feels human.

This app is built around one simple idea: being heard. That is exactly where most similar applications fall short. Not everyone can pick up on the quiet, unspoken emotions behind a message, and without that sensitivity, the support feels hollow.

My approach goes beyond simply plugging in an OpenAI API and calling it a day. I draw upon my professional background as a certified TESOL/TEFL educator and my specialized training in "Teaching Math for Understanding" from UCSC to bridge the gap between technical architecture and genuine pedagogical support.

Instead of generic responses, I am building a series of service layers that act as emotional filters. These layers draw on Jung's cognitive theory to help the system recognize not just what a user is saying but how they might be feeling. By combining my teaching expertise with these psychological frameworks, the system can respond in a way that truly meets students where they are, providing guidance that is both intellectually structured and emotionally resonant.

STACK CHOICES

- Back-end: Laravel (10/11)
- Front-end: Mostly Vue3 and Bootstrap 5 (customized)
- Database: MySQL

TECHNICAL IMPLEMENTATION

Due to page limitations, this description is highly conceptual. More detailed considerations will be implemented in Phase 2.

Design Language & Theme

The application's visual identity is inspired by my own creative project, a warm, teddy bear-themed design system (referenced at exam.bearbei.com). This playful yet approachable aesthetic is intentional. Given that the target users are high-stress students, the design aims to create a welcoming and emotionally comforting experience, reducing anxiety and encouraging engagement with the counseling features.

Front-end: Vue 3 (Composition API) & PWA

The front-end will be built using Vue 3 with the Composition API, prioritizing a lightweight and portable architecture. To maintain this efficiency, I am moving away from heavy reliance on Blade templates, opting instead to build the interface primarily through small, reusable Vue components. This strategy ensures the codebase remains highly modular, performant, and easy to maintain.

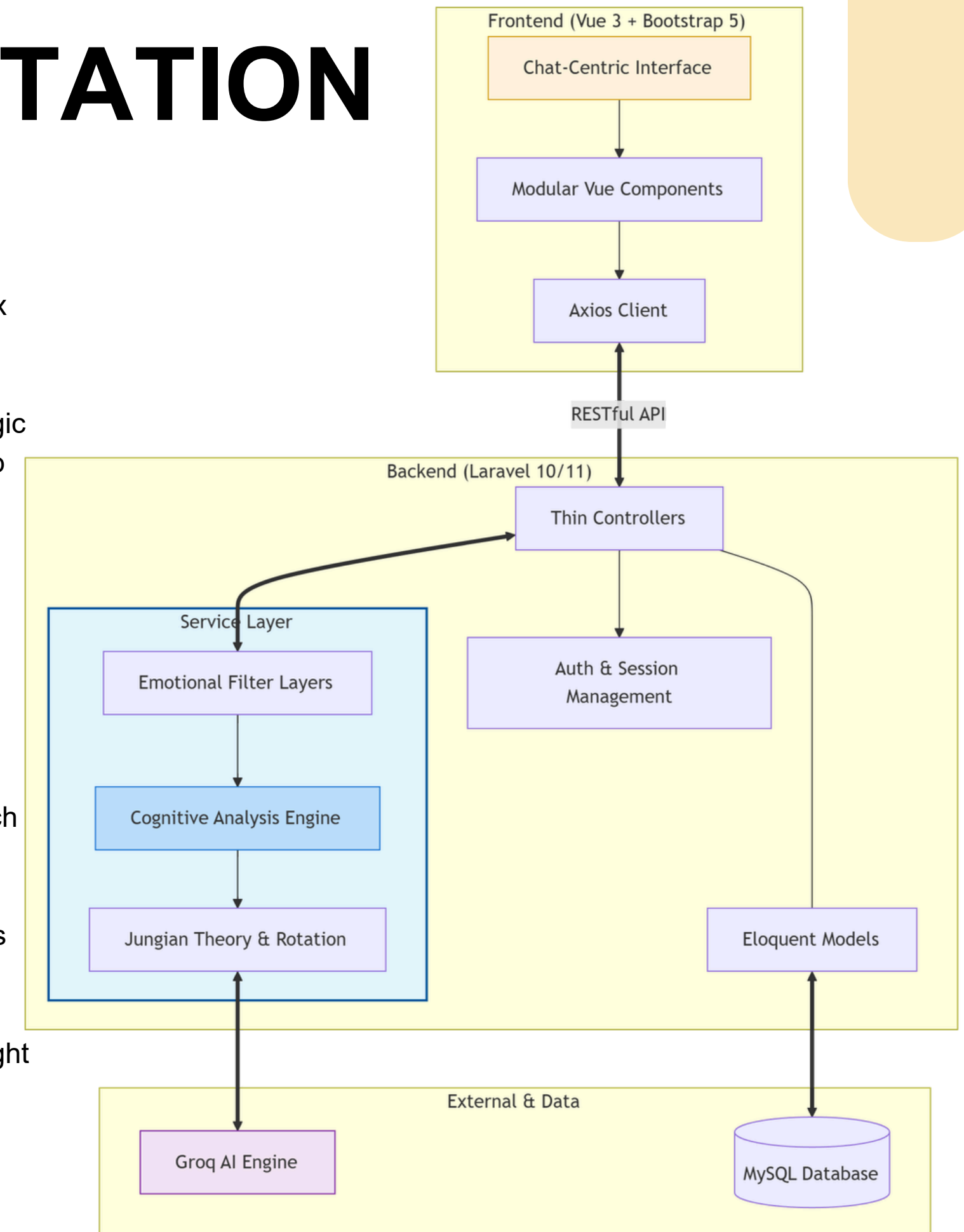
- Chat-Centric Interface: The UI will follow a chat-style interface, providing a conversational and intuitive experience for users seeking counseling support.
- Asynchronous Communication: Components will communicate with the back-end via Axios, adhering to RESTful API conventions for all data exchange.
- Responsive Design: Bootstrap 5 will be used for styling and layout, ensuring a polished interface and seamless responsiveness across all screen sizes and devices.

TECHNICAL IMPLEMENTATION

Back-end: Laravel (PHP)

The back-end will be built using Laravel, following a structured layered architecture to ensure maintainability and scalability.

- **Models:** Defined for each database entity, these will handle data representation and complex relationships.
- **Thin Controllers and Service Layer:** Controllers will remain lean by delegating business logic to dedicated Service classes. This modular approach makes individual services easy to swap or extend in future iterations.
- **Session Management:** Used to handle temporary state and in-progress calculations during user interactions.
- **Authentication:** A robust system will be implemented to manage access control and ensure that all user data is persisted securely.
- **Groq Integration:** We will leverage a dedicated Laravel package to interface with Groq, which serves as the application's central cognitive engine.
- **Cognitive Analysis Engine:** The service layer will perform real-time analysis of the student's state of mind. By identifying which of the eight cognitive functions the user is currently employing, or detecting specific patterns like the 1-3 function loop, the system will apply cognitive rotation techniques. These interventions are designed to disrupt unproductive thought patterns and guide the student back to a state of calm. Further technical specifications regarding the application of Jungian cognitive theory and rotation methods will be detailed in the project's README file.



Overview of the Architecture diagram